

Candidate Name

Centre Number

Candidate Number

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For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS
PAPER 2

6032/2
1 hour 30 minutes

SPECIMEN PAPER

Additional materials:
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

For numerical answers, **all** working should be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

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1	
2	
3	
4	
5	
6	
TOTAL	

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DATA

speed of light in free space	$c = 3.00 \times 10^8 \text{ ms}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$ ($1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ mF}^{-1}$)
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant	$h = 6.63 \times 10^{-34} \text{ Js}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	$R = 8.31 \text{ JK}^{-1}\text{mol}^{-1}$
the Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
acceleration of free fall	$g = 9.81 \text{ ms}^{-2}$

FORMULAE

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p \Delta V$
gravitational potential	$\phi = - Gm/r$
hydrostatic pressure	$p = \rho gh$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_o \cos \omega t$
	$v = \pm \omega \sqrt{(x_o^2 - x^2)}$
Attenuation of x-rays	$I = I_o e^{-\mu x}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
energy of charged capacitor	$W = \frac{1}{2} QV$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \dots$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_o \sin \omega t$
radioactive decay	$x = x_o \exp(-\lambda t)$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$

1 (a) (i) Define

1. *centripetal force*,

2. *centripetal acceleration*.

[2]

(ii) State the direction of displacement of an object moving in a circular path.

[1]

(iii) Explain why centripetal force does not do work on an object moving in a circular path.

[2]

- (b) A hunter launches a spear from a height of 3.0 m above the ground with an initial speed, u , of 25 m/s. The spear is thrown at an angle of 60° to the vertical as shown in Fig. 1.1.

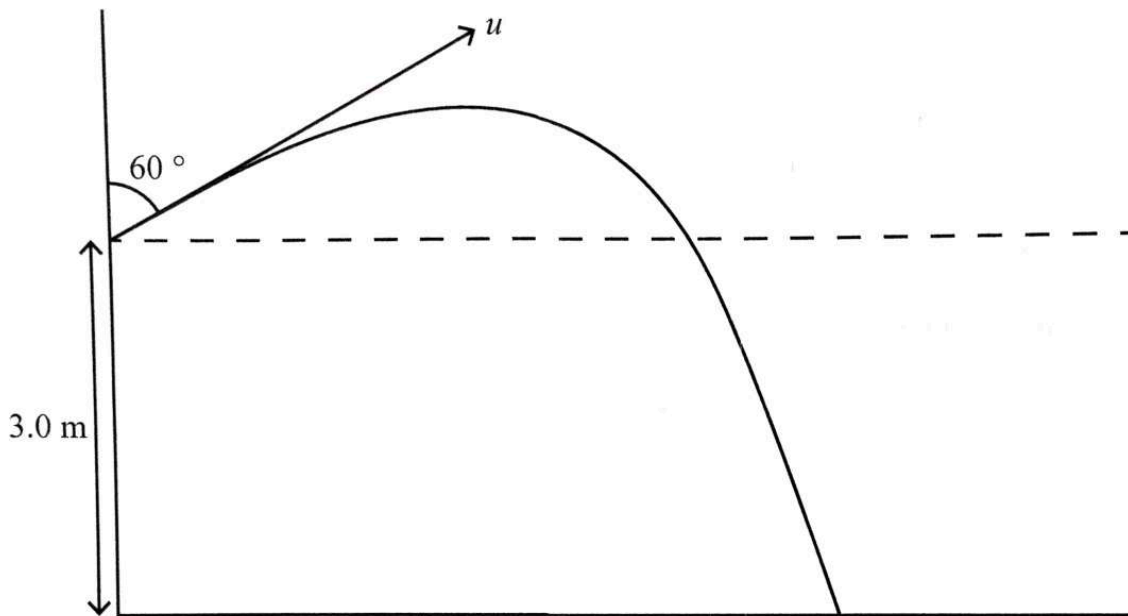


Fig. 1.1

Calculate the

- (i) time of flight until it hits the ground,
- (ii) speed of the spear at the instant it strikes the ground.

[5]

2

(a)

State the **two** conditions that must be satisfied for a diffraction maxima to exist.

[2]

(b)

A parallel beam of monochromatic light of wavelength 650 nm is directed normally at a diffraction grating which has 600 lines per mm.

Determine the

(i)

number of diffracted beams,

(ii)

angle of diffraction of the highest order.

[5]

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- (c) An arrangement that is used to determine the speed of sound in air is shown in Fig. 2.1.

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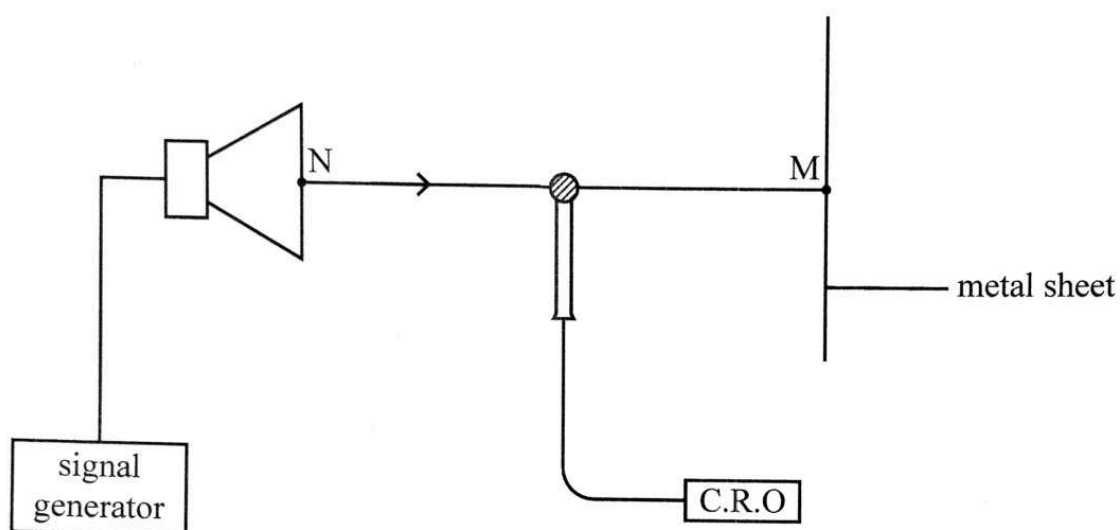


Fig. 2.1

Sound waves of constant frequency are emitted from the speaker N. Describe the observations on the CRO display as the microphone is moved along MN.

[3]

3

(a)

A Physics teacher took learners to the play centre to demonstrate simple harmonic motion using swings. Inextensible cables of length, l , are attached to a horizontal bar and experience no frictional force at pivot. **Fig. 3.1** shows the arrangement.

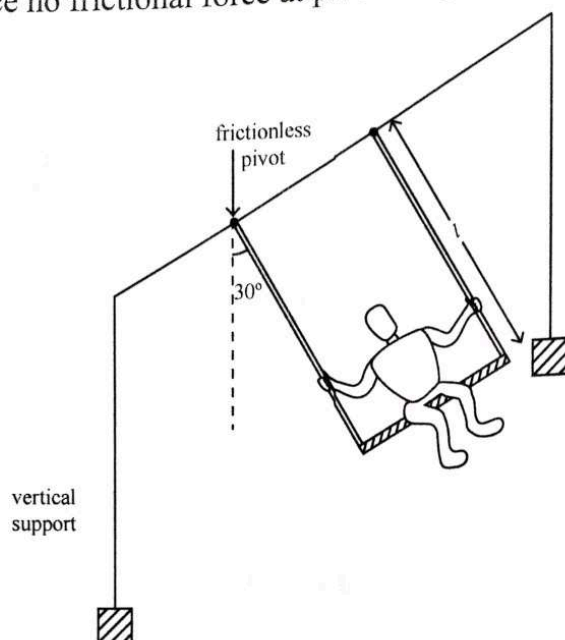


Fig. 3.1

A learner of mass 50 kg rides the swings which is assumed to execute simple harmonic motion. The period of the swing is 2.5 s.

Calculate the

- (i) length of the cable,
- (ii) potential energy if the swing is displaced through 30° to the vertical.

[4]

(b) Describe how the teacher can demonstrate using the swings

(i) heavy damping,

(ii) resonance.

[4]

(c) Organisers of a traditional dance ceremony designed a musical instrument which can produce variable sound from a buzzer. A jockey is made to slide along a 1 m long wire AB of constant resistance per unit length as shown in Fig. 3.2.

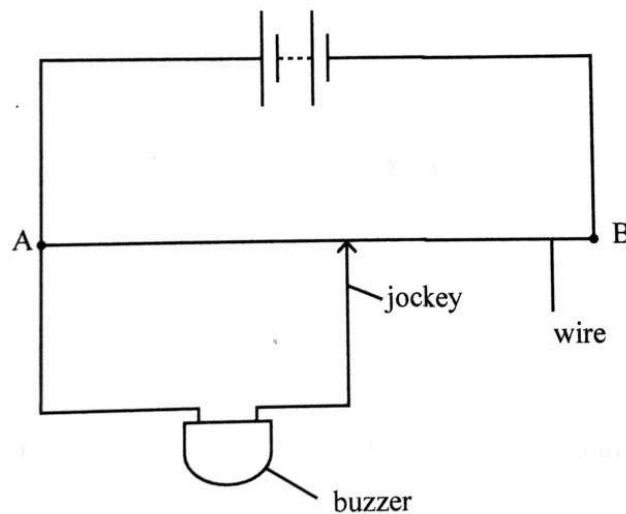


Fig. 3.2

Explain how a variable output is produced by the buzzer.

[2]

- (a) A farmer designed a logic circuit that provide heat and light to the chicken project when necessary. The circuit is composed of a dc source, a high resistance bulb, and switched S_1 and S_2 as shown on Fig. 4.1.

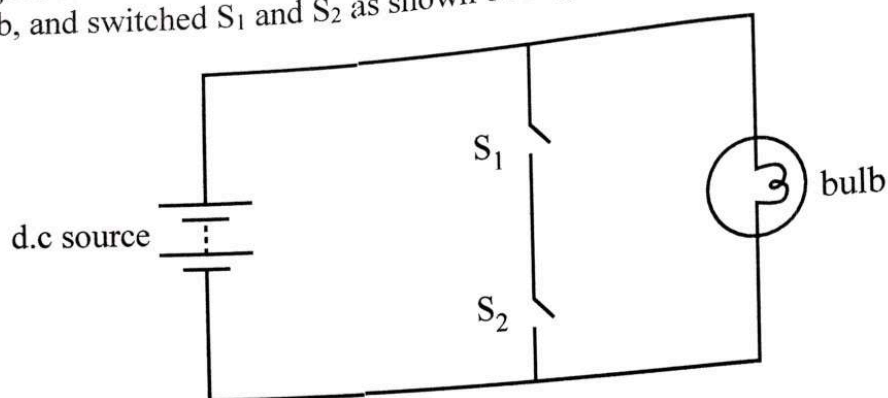


Fig. 4.1

- (i) Draw the truth table for the logic circuit.

[Switch open, 0 bulb on 1
Switch closed 1 bulb off 0]

- (ii) Identify a logic gate that operates as the logic circuit.

[3]

- (b) In a robot designed to prevent collisions, sensors are used.

- (i) Identify any type of sensor that may be used to perform this function.

- (ii) Explain how the sensor works in preventing collisions.

(c) Explain the use of transformer oil in terms of its

(i) low viscosity,

(ii) high thermal conductivity,

(iii) low acidity.

[3]

5 (a) Explain using the Zeroth law of thermodynamics why traditional methods are less suitable to determine the temperature of the body.

[2]

(b) In laboratories, accurate temperature measurement is the key, from open flames used in experiments to biomedical research involving ultrasound.

Suggest a suitable type of thermometer to measure

(i) the temperature of a Bunsen burner flame.

[1]

- (ii) the change in temperature of small crystals under ultrasound pulses,

[1]

- (iii) a small change in human body temperature.

[1]

- (c) An ideal gas at 100°C , pressure $4.9 \times 10^5 \text{ Pa}$ and volume $2.4 \times 10^3 \text{ cm}^3$ is enclosed in a cylinder.

Show that the number of molecules of the gas is 2.3×10^{23} .

- (d) A thermometer was used to measure the temperature of oil.

Determine the temperature change of the oil.

Given that: mass of oil 32.0 g
 specific heat capacity of oil $140 \text{ Jg}^{-1}\text{K}^{-1}$
 initial temperature of oil 54.0°C
 mass of thermometer 12.0 g
 specific heat capacity of thermometer $0.180 \text{ Jg}^{-1}\text{K}^{-1}$
 initial temperature of thermometer 19.0°C

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- 6 (a) State **two** pieces of evidence provided by the photoelectric effect for the particulate nature of electromagnetic radiation.

[2]

- (b) The three lowest energy levels of the electron in the hydrogen atom have energies:

$$\begin{aligned} E_1 &= -21.8 \times 10^{-19} \text{J} \\ E_2 &= -5.44 \times 10^{-19} \text{J} \\ E_3 &= -2.42 \times 10^{-19} \text{J} \end{aligned}$$

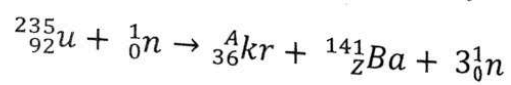
- (i) Explain why the energy values are negative.

[1]

- (ii) Calculate the wavelength of the hydrogen spectral line due to the transition levels from E_2 to E_3 .

[2]

- (c) A typical nuclear fission is represented by the equation:



Deduce the value of

- (i) A, _____
- (ii) Z, _____

[2]

- (d) Use the data to calculate the binding energy per nucleon in MeV for an alpha particle.

proton mass	=	1.0073 u
neutron mass	=	1.0073 u
mass of ${}_2^4\text{He}$	=	4.0015 u

[3]

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